PressLens
Work Chat Context

Researcher-led | Clean-slate Analytics Cup 2026 | Repository-ready handoff

Important scope note: This is a snapshot of the repository starter and handoff prepared on November 2, 2026. It is **not proof** that GitHub access works, the data were audited, or the methodology was empirically validated. The Work chat must verify real data access before beginning any experiments. Continue using SkillCorner competition-eligible datasets only.

Included sections: 00 — OPENING INSTRUCTIONS, 01 — START HERE, 02 — RESEARCH STATE, 03 — SCIENTIFIC RESEARCH BRIEF, 04 — RESEARCH WORKFLOW, 05 — DATA AUDIT PLAN, 06 — FEASIBILITY MATRIX, 07 — RESEARCH INTEGRITY PROTOCOL, 08 — FOCUSED LITERATURE REVIEW, 09 — SOURCE REGISTER

00 — OPENING INSTRUCTIONS

Original: WORK_CHAT_START.md

PressLens — Start New ChatGPT Work Execution Chat

Version: 1.1 · **Date:** 2026-11-02 · **Purpose:** move the approved clean-slate scientific framework into execution now that the new Mac repository starter has been installed. This is a *handoff instruction*, not empirical evidence.

I have set up a new clean-slate PressLens repository on my Mac, using the repository starter produced for this project. The **expected local location** is `~/Documents/GitHub/PressLens-Cup-2026` and the **intended GitHub remote** is `FHaddad99/PressLens-Cup-2026`. **Verify the actual local path, Git branch, commit, remote URL, working-tree status and accessibility; do not assume the GitHub remote was successfully created or pushed.** The repository is private by default if it was created using the installer. If GitHub access is unavailable to you, check whether the desktop Work environment has approved local-folder access or use the attached context files. Do not claim to have inspected a repository or files unless you actually read them.

Purpose and scientific direction

We are conducting a clean-slate, **researcher-led SkillCorner × PySport Analytics Cup 2.0 football** project. I am the scientific principal investigator; you are the transparent technical/scientific implementation partner. I prefer natural speech-to-text discussion and opportunities to intervene **before** substantial methodology choices. Do not use Codex, import the old Codex PressLens repo, or inherit old gate/model/audit conclusions as verified evidence.

Our central question is whether coordinated presses reduce the **feasibility and attacking value of the options that survive**, including the indirect contributions of off-ball defenders who record no tackle or interception. The importance of a press is not the number of lanes removed or the eventual outcome alone. My football reasoning is documented in `docs/RESEARCH_BRIEF.md`: (1) Press B leaving several harmless choices beats Press A leaving one clear dangerous one; (2) a lone defender splitting a two-on-one trades space, angles and time to manage risk; (3) delay can allow recovering teammates to suppress future opportunities; and (4) individual and collective credit must avoid double-counting. Distinguish option feasibility, action-selection probability, action danger/value, measured change during a press, defender-attributable counterfactual influence, and realized results. Do not assert access to players' mental decision trees or causal findings without defensible evidence.

Initial research scope: organized pressing, supporting off-ball defenders, and within-press containment. Counterpressing only if data support it. General defensive transitions are outside the first study. Core: team-level time-resolved opportunity feasibility/danger with appropriate simple baselines. Conditional: individual off-ball contributions. Stretch: delayed credit, pose, complex selection models. The exact method is **not approved**; adapt after evaluating the data. A small ground-pass feasibility/value experiment is a *candidate next milestone*, not yet authorized.

Read the actual repository before acting

Read, in order:

1. `START_HERE.md`
2. `RESEARCH_STATE.md` — current live checkpoint, which may have changed since this handoff

3. `docs/RESEARCH_BRIEF.md`
4. `docs/WORKFLOW.md`
5. `docs/data/AUDIT_PLAN.md` and `docs/data/FEASIBILITY_MATRIX.md`
6. `docs/research/RESEARCH_INTEGRITY_PROTOCOL.md` and `docs/research/VERIFIED_SOURCE_REGISTER.csv`
7. Relevant code, including `tools/inspect_source.py`, only as required for the first task.

Prefer the **actual repository files** over the attached fallback snapshots if they differ; report conflicts to me rather than silently reconciling them. The repo scaffold and synthetic tests do not count as a completed data audit, validated model, or live GitHub confirmation.

First executable milestone — data discovery and feasibility, not model development

1. Confirm what you can **actually access**: the new PressLens repository and the eligible SkillCorner source checkout/folders on this Mac. The SkillCorner source should be accessed **read-only** and kept outside the repository. Do not assume data are available based on documentation or the old Windows path.
2. Verify official competition data rules and inspect the official SkillCorner resources. Use only eligible SkillCorner competition datasets for match/event/tracking/pose training and empirical evaluation. Existing published literature and appropriately licensed source code may inform methodology, but do not import external match datasets or assume externally fitted weights are eligible. Do not publish raw/restricted data or unapproved derivatives.
3. Perform the **bounded metadata/schema/source inventory** authorized in `docs/data/AUDIT_PLAN.md`. Distinguish documented fields from locally confirmed fields and sampled coverage from full-file verification. Show me early findings in plain language before expensive or broad full-file scans.
4. Then propose the minimum **targeted quality and synchronization audit** necessary to test which passing-opportunity models the data support. Explain which questions this audit can actually decide, its stopping rule, and where my tactical/scientific judgment matters.
5. Present a human-readable data dictionary, a method-feasibility matrix, usable sample counts with denominators/flags when measured, limitations, and a single proposed small experiment with comparison baselines and pass/fail/abstention criteria. **Obtain my approval before implementing the passing-feasibility/value model.**

Non-negotiable scientific and operational safeguards

- Do not invent publications, DOI strings, results, timestamps, sample sizes, field definitions, trajectories, test evidence, or performance advantages. Verify bibliographic identity **and each used claim** against primary original sources; treat publisher records, author-reported results and our reproduced results separately. Provider-derived SkillCorner metrics are not independent ground truth.
- Keep every significant methodological choice understandable to me **before** implementation; routine coding details can be delegated. Perform bounded work, show actual file paths, commands, success/failures, and what remains uncertain. No invisible multi-gate or broad side-project execution.
- Keep original observations external to the Git repo, local/private reports ignored, and do not publish source data. Check `git status` before modifying anything and do not overwrite existing changes.
- Do not restart audits endlessly: audit only until the next experiment can be responsibly scoped. If a limitation is manageable, document it or abstain; do not change thresholds to manufacture success. Preserve negative results and maintain evaluation separation where feasible.

- Update `RESEARCH_STATE.md` after actual work and use it as the single continuity checkpoint. Separate **software correctness / empirical evidence / football usefulness / causal claims**.
- Competition deliverable: an understandable, reproducible open-source method/tool, concise competition README and one-minute pitch; there is also a parallel basketball entry. Target submission before **2026-12-18**, with the final organizer cutoff verified.

First reply requested

Please begin by telling me: **(A)** what repository/data access you actually verified, **(B)** what stage the live `RESEARCH_STATE.md` shows, **(C)** a short, bounded plan for the first metadata/schema audit and its stopping point, and **(D)** any one essential access request you need. Then carry out only the authorized initial inventory where access exists, showing real evidence as you go.

At each checkpoint always finish with: **CURRENT STAGE | COMPLETED | VERIFIED EVIDENCE | LIMITATIONS | MY DECISION NEEDED | ONE NEXT ACTION**.

01 — START HERE

Original: START_HERE.md

START HERE — one-screen operating guide

1. Read RESEARCH_STATE.md to see the **actual current stage**, verified work, open decisions, and **exactly one next task**.
2. Read docs/RESEARCH_BRIEF.md for the football reasoning and scope approved on 2026-11-02. **The mathematical implementation is not frozen.**
3. Consult docs/WORKFLOW.md before Work starts a substantive task. The user is principal researcher; ChatGPT Work performs bounded, visible tasks and returns for scientific decisions.
4. For the immediate milestone, inspect docs/data/AUDIT_PLAN.md and docs/data/FEASIBILITY_MATRIX.md. Do **not** build the passing model before data feasibility is reviewed.
5. Use docs/research/RESEARCH_INTEGRITY_PROTOCOL.md and docs/research/VERIFIED_SOURCE_REGISTER.csv as research-integrity controls. Registered sources are leads: verify specific claims at primary sources when using them.
6. Keep SkillCorner raw files **outside this repository**. Private reports stay ignored. No GitHub release until data/report publication rights are checked.

New Work chat opening: "Read START_HERE.md, RESEARCH_STATE.md and docs/RESEARCH_BRIEF.md. Confirm which SkillCorner data you can actually access. Explain and execute only the bounded discovery/feasibility audit, returning with a human-readable decision before any model implementation."

02 — RESEARCH STATE

Original: RESEARCH_STATE.md

PressLens — Live research checkpoint

As of: 2026-11-02 (repository starter)
Stage: 0 — new workspace and data-access verification
Owner-approved: scientific objective and prioritization, NOT a final measurement methodology
Empirical status: NO clean-slate SkillCorner audit, experiment, validation or trained models reported.

What is established

- Football focus: organized pressing, support defenders and within-press containment; counterpress only if feasible, general open-field transition out of initial scope.
- Primary idea: feasible *and* dangerous surviving options matter more than number of eliminated options; track them over time.
- Distinguish immediate suppression, delayed enabling value, observed outcomes, and counterfactual individual attribution.
- Priority: team-level ground-pass opportunity measurement first **if the data support it**; individual credit conditional, delay/pose stretch.
- Dataset restriction: eligible SkillCorner competition materials only; external literature for methods, not extra matches or automatically eligible weights.
- Research integrity: primary evidence, explicit uncertainty and traceable experiments.

Actual implementation and verification

- ☑ Source repository starter, baseline data-discovery CLI, synthetic tests, Git safety checks prepared.
- ☐ GitHub remote created/pushed (only mark after verified).
- ☐ SkillCorner local files accessible and official data version pinned.
- ☐ Actual file/schema/field inventory performed on eligible match files.
- ☐ Quality and temporal/coordinate audit performed on a bounded real subset.
- ☐ Initial opportunity-feasibility methodology selected with owner.
- ☐ Any new empirical model executed or validated.
- ☐ Any individual defender attribution performed.

Known unknowns / don't infer

Actual local datasets and coverage; official versions of CSVs/tracking; usable event↔tracking joins; quality of player and ball positions in candidate pressing situations; whether attacking-value calibration is possible from eligible samples; data/derivative publication permissions.

ONE next executable task

Owner + ChatGPT Work: Confirm access to the official SkillCorner competition data checkout (or upload the relevant files securely); run a bounded **metadata/schema inventory** and inspect the official documentation. Explain findings in plain language and propose the minimum deeper audit before doing it. **Stop before model development for owner review.**

Decisions awaiting owner

- Data source location/access and canonical source revision.
- Preliminary press inclusion and sample-quality criteria informed by actual fields.
- Which passing-feasibility baseline to prototype after sufficient audit evidence.

Session update format (keep concise)

Date · stage · work genuinely completed · commands/input versions and counts · evidence file paths · failures and limitations · owner decision needed · **one next task**. Never claim a step passed because tests completed; software and scientific validity are distinct.

03 — SCIENTIFIC RESEARCH BRIEF

Original: RESEARCH_BRIEF.md

PressLens — Clean-Slate Scientific Execution Handoff

Version: 1.0 | **Prepared:** 2026-11-02 | **Status:** Owner-approved conceptual direction; methodological details are **provisional**, pending inspection of eligible SkillCorner data. This document is an execution handoff, **not** a claim that a new implementation, dataset audit, or experiment has already been performed.

0. How to use this handoff

This file is the starting reference for a **new researcher-led ChatGPT Work conversation**, not instructions to resume the former Codex workflow. Treat previously generated Codex code, audit reports, status gates and model claims as archival, NOT automatically validated or incorporated. The prior repository may remain intact for historical reference. The new Work conversation should show completed work and empirical evidence as it goes. Do not accept a previous gate result by inheritance.

Owner's central requirement: understands and can defend **every meaningful scientific and tactical decision end to end**; natural-language or speech-to-text conversations before implementation; minimal black-box autonomous work. Work performs only a bounded, transparent task, explains and shows evidence, then seeks researcher input before substantive methodological choices. Routine engineering choices need not be micromanaged.

1. Research objective in the owner's own conceptual terms

PressLens aims to quantify the **value of coordinated defensive pressing**, including off-ball defenders who restrict viable attacking options **without recording a tackle, interception, or on-ball pressure action**. The defense's success should be measured by how it reduces the **feasible danger of the attacking options that survive**, not by how many lanes it removes, possession regain alone, or how aggressively the nearest defender presses the carrier.

Research question (provisional): Can tracking-based measures quantify how the feasible and valuable passing/carrying opportunities available to an attacking team evolve during pressing, and can some of that change be attributed defensibly to individual on- and off-ball defenders?

Owner's football reasoning (important context):

- **Press A vs B:** One defense suppresses three routes yet leaves one extremely dangerous easy route; another leaves four harmless backward/sideways routes. The second is clearly superior. The **danger and feasibility of surviving actions** matter more than count of options eliminated.
- **Two-on-one defender:** An intelligent lone defender splits the gap rather than diving at the ball. Angle of approach, velocity, positioning, space and time constrain both direct carry and passing lane. Delay can allow teammates to recover. Defending is an optimization of opportunity **and time**, not simply proximity to carrier.
- **Hockey penalty kill analogy:** Defenders distribute attention according to **risk**, avoiding a single open high-danger chance; this does not imply equal coverage of every option.
- **Defensive delay:** A containing defender can create **future** defensive value by buying time for supporting teammates to track back, even without immediate reduction in passing-lane threat. Those

recovering defenders contribute too. The initiating defender may be critically important but no automatic rule assigns them more credit. Interaction-based attribution must avoid double-counting.

- **Execution vs outcome:** If defenders force a low-probability attacking action and the attacker succeeds through exceptional execution, the defense may have executed well. Report opportunity reduction separately from realized turnover/escape/shot outcomes.
- **Option feasibility, selection, and attacking value** are separate constructs. Tracking-based estimates cannot reveal exactly which alternatives a carrier mentally considered. Do not assert mental decisions or causal effects from observational data alone.

Interdependent factors: options, time and space should be modeled jointly as appropriate; beware double-counting time already embedded in feasibility. Prefer a continuous value measurement before any arbitrary binary success threshold.

2. Initial scope and priority ordering

Primary scope: organized pressing against an opponent in possession, with supporting defenders and within-press containing/delay behaviors; consider counterpressing only if identifiable and data allow. General open-field transitions and pure delay defense are outside initial scope. Include pressing/carrying actions in the conceptual question, but **pilot ground passes first if the eligible data make those easier to measure reliably**. Do not imply a passing-only proof covers all carries.

1. **Core:** verified reconstruction of the dynamic attacking opportunity landscape (eligible passes, limited carry proxy if justified), feasibility, danger, and change over pressing episodes, compared to simple baselines.
2. **Conditional:** individualized off-ball contribution estimates and counterfactuals, **only** if team/option measurement and assumptions support them. Already-published methods study off-ball influence; do not claim invention of the general concept.
3. **Stretch:** future-value of delay/recovery, interaction-aware credit, choice-weighted action selection, contextual norm baselines, and pose-enhanced feasibility. Pose is never a prerequisite.

Do not freeze methods before seeing actual source coverage. Preserve scientific integrity and experimental separation while adapting models or narrowing claims to match data quality. An optimal method is not known upfront. Do not change definitions or thresholds merely to improve favorable results, and preserve failed and alternative model results.

3. Mathematical constructs to investigate, not yet adopted

- For proposed action `a` in observable game state `s`, candidate value: `Q(a,s) = p_success(a,s) * V_success(a,s) + (1-p_success(a,s)) * V_failure(a,s)`. This is a conceptual decomposition, **not** a fitted or calibrated SkillCorner model.
- Consider maximum surviving value `max_a Q(a,s)` versus action-selection-weighted expected value `sum_a P(select a|s) Q(a,s)` versus another justified risk-sensitive aggregation. **Do not simply sum values of mutually exclusive actions**; the maximum can ignore having multiple strong options, while selection weights may require training data not available here.
- Distinguish (i) **observed modelled opportunity change over time**, (ii) **estimated defender-attributable influence** requiring a counterfactual, and (iii) **realized match outcome**. They are not equivalent.
- A hand-picked xT proxy is a **heuristic score** unless calibrated on eligible data and externally validated; do not call it an empirically calibrated expected value.

- Counterfactuals such as removing a defender may be physically or tactically unrealistic. Any attribution is conditional upon explicit assumptions and must not automatically add interaction credit multiple times.

4. Where the literature places the novelty boundary

Original studies/records to read and verify for specific claims before they appear in a final submission:

- Fernández, Bornn & Cervone (2021), expected possession value. DOI: https://doi.org/10.1007/s10994-021-05989-6. Rich pass/drive/shot action/outcome/choice formulation; data and training demands may exceed current eligible dataset.
- Bischofberger & Baca (2026), Dangerous Accessible Space. DOI: https://doi.org/10.1186/s40537-026-01387-8 ; code https://github.com/jonas-bischofberger/accessible-space. Physics-based hypothetical passes, accessibility and defender counterfactuals **already** address off-ball suppression. Limitations include selection probabilities, low/high passes, realistic counterfactual assumptions, parameter portability and computation.
- Everett et al. (2025), GAPP graph-attention off-ball defensive influence. DOI: https://doi.org/10.1109/DSAA65442.2025.11248003. More demanding training sample; attention-based influence is not automatically causal.
- Kim et al. (2026), DEFCON preventive defender value. https://arxiv.org/abs/2512.10355. Original Sloan paper and code available. Uses external training data; not automatically competition compliant.
- Arbués-Sangüesa et al. (2020), orientation and pass feasibility, https://doi.org/10.1109/CVPRW50498.2020.00451. Pose has only two eligible matches, and receiver prediction is not completion calibration.
- Bojinov & Bornn (2016), defensive disruption, https://www.sloansportsconference.com/research-papers/the-pressing-game-optimal-defensive-disruption-in-soccer.
- The alleged 2026 **Pass Suppression Value** thesis from secondary indexing remains **pending original university-record verification**. Do not cite numerical findings from unverified thesis secondary summaries.

Preliminary, not proven novelty opportunity: an interpretable, press-specific and **time-resolved** analysis of the **feasible, valuable surviving actions**, with explicit quality flags, a measurable gain beyond simple pressing/geometry baselines, and conditional off-ball credit. Neither temporal pressure nor off-ball suppression in general is novel; demonstrate the **specific difference** or report an applied/OSS contribution honestly. Literature references are methodological background, not extra match/training data.

Companion evidence files in this package are **source leads**, not substitute for opening primary material. Distinguish publisher-confirmed bibliography, author-reported empirical claims, and our own reproduced results.

5. Competition rules and permitted data (verified from official PySport page as of 2026-11-02)

Official rules: https://pysport.org/analytics-cup/editions/analytics-cup2 ; official SkillCorner resources: https://skillcorner.github.io/opendata/ and https://github.com/SkillCorner/opendata.

- Deadline **December 18, 2026** (verify organizer's final timezone/portal). User also intends a **separate basketball submission**, which must not be sacrificed.

- **Only SkillCorner data supplied for Analytics Cup 2.0 and/or previous Analytics Cups may be used as empirical inputs.** No outside football match/event/tracking, video, or training datasets. Inspect any model/package **pretrained parameters or weights**; existing licensed code may be reused/disclosed, but that does not automatically make external fitted data/weights eligible. Obtain organizer clarification if ambiguous.
- Official overview: football 20 XY matches (including 10 new), dynamic events on 10 matches, body pose for 2 matches, season aggregates of physical measures across 175 matches, tutorials/documentation/viewers. **These are advertised resources, not proof of actual local file completeness.** Match metadata, frames, field transforms and tracking quality must be checked. Provider documentation may emphasize the original ten-match subset; reconcile actual official sources.
- Submission: **one-minute or shorter self-presented/self-voiced YouTube pitch**, **public GitHub repo** of open-source code and relevant project resources **without the datasets**, properly licensed including LICENSE.md, **README <=1,000 words and maximum two figures/tables combined** with clean-environment reproduction instructions. No proprietary software required. Projects must contain substantial original work and remain open-source.
- Evaluation criteria equal **20% each:** relevance, methodology, originality, communication and open-source potential. A traditional academic manuscript is **not required**; build a scientifically defensible, intelligible tool or method with transparent evidence and limits.
- One football entry and one separate basketball entry allowed in same chosen region; region/final attendance requirements apply. Recheck live official site and competition Discord for changes before submitting.

Rule of separation: external *ideas, papers, open-source code* may be read/adapted under the rules. Additional *match observations, empirical training labels, cached training examples, video, fitted weights/parameters trained on external match data* are not implicitly permitted. Do not publish official restricted datasets or unapproved derivatives.

6. Data types to inspect, not assume usable

- **XY tracking**: approximately 10 Hz; frame/timestamp/period, player and ball coordinates, player/possession identity, detected vs extrapolated positions, broadcast field-of-view coverage.
- **Match metadata**: lineups, player identities, playing time and pitch geometry.
- **Dynamic events** (advertised ten matches): pressure/engagement and possession metrics; possible provider-generated `passing_option_ease`, `overall_pressure`, `time_to_impact`, `possession_epv` fields. Audit actual schema, subtypes, units, reference times and event alignment rather than assuming any field applies to each event/frame. Provider metrics are not independent ground truth.
- **Phases of play**: in/out of possession phase labels and frame boundaries, possibly mismatched times and period references.
- **Pose**: 29 joint estimates at 25 Hz for two matches; model-predicted joint errors, detection-only availability, 2D alignment limitations; optional exploratory ablation.
- **Season aggregates**: not a substitute for frame-wise attacking decision labels or pass-selection training data.

First authorized execution milestone: source inventory, targeted audit and feasibility map

Before any passing-feasibility experiment, in the new Work chat:

1. **Confirm data access and official competition sources**; ask for the appropriate local SkillCorner data folder permission in desktop Work, or an uploaded/mounted dataset where necessary. Do not claim access to the old Windows path from cloud/mobile Work. Preserve old provider checkout read-only; create new output location. Never pull non-eligible games.
2. **Inventory all competition-eligible resources and documentation**: match IDs and files, tracking and events/phase coverage, metadata, pose, aggregates, viewers, schema PDFs/tutorials/loaders, versions/licenses, available fields. Record both provider-documented and actually inspected resources.
3. **Build plain-English data dictionary and field usage map**: identities, frame/period/time semantics, coordinate units/transformations, observed versus predicted/provider-derived metrics, join keys and relevant fields for press identification and passing action value.
4. **Targeted integrity/quality checks**, based on actual eligible files: missing player/ball tracks, detected vs extrapolated rates, discontinuity, unrealistic velocities, off-camera players, identity/possession mismatch, event/phase join exceptions, duplicate/absent keys. Distinguish complete match audit from sample inspection and flag unknown full-file coverage. Report numerator and denominator for all measured rates.
5. **Press-specific feasibility check** on a bounded, actual verified subset. Are enough reliable carrier/receiver/defender trajectories, ball positions and aligned pressure episode labels available for a simple pass feasibility experiment? If not, transparently define a fallback.
6. **Stop and present a human-readable decision packet** before implementation: complete resource inventory; useful/unknown/broken fields; clear visual of one real SkillCorner scenario *if available*; candidate model alternatives; limitations; one scoped experiment with pass/fail/abstain conditions and exact recommended data subset. Owner approves the method next.

Stop criteria: dataset/field provenance and temporal/spatial semantics are sufficiently understood for a clearly defined subset; quality/exclusion rates are measured, baseline implementable, unknowns bounded. This is not permission to spend days resolving issues irrelevant to the first experiment.

If data cannot be accessed: report the precise missing file or permission; provide one minimal request to user. Do not fabricate audit results or substitute third-party matches.

7. Next experiment — purpose, but NOT YET AUTHORIZED

Purpose: test whether actual SkillCorner frame-level states can support meaningful estimates of the feasible and attacking-value-weighted **potential ground passes**, and whether these change during pressing episodes. This is a **component feasibility test**, not proof of pressing value or defender causality.

Candidate comparisons: geometric proximity/lane baseline; simple motion/time-to-interception feasibility; feasibility + honest value score. Define pass candidate universe, ball trajectory assumptions, receiver motion, tracking quality exclusions and action value source *after data audit*. Reserve separate review/evaluation examples where sample allows. Observed passes can check selected-action success; unchosen hypothetical passes cannot be directly labelled counterfactual success from match records. Differentiate software unit tests, statistical discrimination/calibration, tactical usefulness and causal interpretation. Pitch-based visual review preferred to tables-only.

Potential adaptation: if velocities unreliable, fallback to geometry; if calibrated value unavailable, label danger as heuristic or test feasibility alone; if press alignment unreliable, narrow to verified episodes.

Deliberately do not skip back into complex EPV/GNN or pose unless the evidence and remaining time justify it.

8. Research ethics and hallucination guardrails

No fabricated citations, authors, DOIs, source claims, sample sizes, training runs, player trajectories, test outputs, novelty claims or conclusion. An existing citation misrepresenting original findings is also an error. Verify identity **and exact claim** against original primary papers/publisher records; source status `P` primary verified, `A` author-reported, `S` secondary only, `H` hypothesis, `U` unverified/exclude, `R` reproduced on our approved sample with code/denominators. Verify paper's sample and evaluation methodology when citing numbers. Preserve negative results.

Use companion `RESEARCH_INTEGRITY_PROTOCOL.md` and `VERIFIED_SOURCE_REGISTER.csv` (provisional leads, subject to fresh source checks). Evidence must be visibly split into **documented**, **computed in this Work session**, **hypothetical**, **unknown**. No false claim that cited author outcomes replicate on SkillCorner data.

9. Researcher-led operating protocol

- **Before each scientifically substantive task:** state objective, why now, required SkillCorner inputs, alternatives, assumptions, likely pitfalls, stopping condition, and any decision the researcher should make; obtain approval before choosing a major model, pressing definition, value function, credit allocation or threshold that changes scientific conclusions.
- **Execute bounded work transparently:** show code/files created, datasets actually read, exact tests and failures, output examples, and scientific caveats; no hidden expansion or parallel 'side quests'. Work may complete routine engineering without repeated permissions.
- **After each task:** concise plain-English `CURRENT STEP | COMPLETED | VERIFIED EVIDENCE | LIMITS/ERRORS | OWNER DECISION | ONE NEXT ACTION`. Never conflate code tests, empirical evaluation, and tactical review. Don't promise asynchronous results unless explicitly using a scheduled task.
- **Review football on a pitch** when possible (show real SkillCorner players/ball/route/time and meaningful uncertainty), not via giant JSON dumps or manual reconstruction. No invented plot data.
- **Revisions are expected**, but record why they improve science, exactly which past comparisons must be reconsidered, and preserve previous failures. Methodology is adaptive; research objective and scientific honesty are stable. Do not endless-audit or endlessly restart.
- **Maintain one concise live progress file** (e.g. `RESEARCH_STATE.md`) plus reproducibility records and source register. After each checkpoint, user can open another Work/Chat conversation using these files; no expectation of an infinite context window.

10. Immediate action for the next Work chat

DO NOT begin model implementation. Open/read handoff and companion integrity documents, confirm you understand the central football construct, the data boundary, and that this is a **clean slate**. Then prepare and undertake the **dataset resource inventory and targeted audit** as the first bounded task, after checking that the eligible SkillCorner data are actually available. Show preliminary inventory and field definitions to the owner before more expensive scans. At the end, prepare a human-understandable feasibility decision packet and request owner approval for the first experiment. Preserve sufficient time for the separate basketball entry.

Success is not a perfect theoretical framework: it is defensible, original-or-usefully-applied, reproducible empirical work that the researcher can explain and the judges can run and understand.

04 — RESEARCH WORKFLOW

Original: WORKFLOW.md

Researcher-led execution workflow (short, not another governance stack)

Role: the owner is scientific principal investigator; ChatGPT Work handles bounded agreed implementation, analysis, code and documentation. Do not delegate to Codex. Speech-to-text scientific discussions are welcome.

Before a substantial step: state the question, why it matters, data prerequisites, plausible alternatives, what could invalidate it, and a short stopping criterion. Obtain owner input before selecting scientifically consequential definitions (e.g., press detection, action value, success criterion, attribution counterfactual).

During: carry out only the authorized step; preserve actual commands, sample sizes, source hashes/revisions, incomplete runs and negative evidence. Show informative, meaningful intermediate findings rather than operating as a black box. Routine engineering changes are fine.

After: report in plain language what actually ran, what the evidence says and does not say, where the user can inspect results (ideally football pitch for tactical review), what decisions remain, and **one next task**. Update `RESEARCH_STATE.md` and the log that changed. No automatic gate PASS.

Scope and change: methods adapt when real data demonstrate limitations. Record revised rationale in `docs/methods/DECISION_LOG.md`, retain previous evaluations and evaluate changes without cherry-picking examples. Defer unrelated improvements; stop audits once the next experiment can be responsibly attempted.

Independent checks: unit tests ≠ empirical predictive evidence ≠ football construct validity ≠ causal attribution. Never infer one from another.

Competition: protect time for the separate basketball entry; keep README, open-source reproducibility and video in scope from the beginning. Public release and publishing derived examples require permission review.

05 — DATA AUDIT PLAN

Original: AUDIT_PLAN.md

SkillCorner discovery and feasibility audit — milestone 1

Why before modelling: the published schema does not prove that local competition files are present, complete, correctly synchronized, or adequate for any proposed measurement. Data suitability determines our method.

Part A — official resource + source pin

- Confirm approved event dataset and competition rules at https://pysport.org/analytics-cup/editions/analytics-cup2 .
- Inspect https://github.com/SkillCorner/opendata and official schema/tutorials; record URL, revision/commit, licence/EULA and which files actually exist locally.
- Inventory **all eligible resource families**: matches/metadata, tracking, events, phases, aggregates, pose where supplied. Distinguish official remote availability from local verified possession. Keep source version and identifiers, not restricted contents, in public logs.
- Produce a field dictionary: provider-defined vs directly observed vs modelled, units, meaning, temporal coverage, join keys, and missingness semantics.
- Start with `python tools/inspect_source.py --provider-root "..."`. That tool inspects file names/headers and tiny samples. It does **not** verify full match records, joins or tracking quality.

Part B — targeted real-data checks (after discussing Part A)

1. Select a small, **documented** subset of candidate organized presses, including negative/nonpress cases if feasible. No outcome-driven cherry-picking.
2. Verify relevant full-match JSONL/CSV parsability, field/value distributions, ball/player coordinates and detection/extrapolation flags on those samples. Report exact denominators and excluded cases.
3. Resolve coordinate orientation/units, time keys, possession/team identities, player playing intervals and event↔tracking alignment. If something is irreconcilable, apply narrow abstention rather than invent coordinates or times.
4. Inventory relevant provider-generated pressure, passing-option and EPV fields. Document provenance and what they can/cannot validate; do not use as independent ground truth.
5. Assess whether meaningful receiver sets, pass geometry, velocities/interception times, and a **clearly labelled** threat/value representation can be supported.

Decisions and exit (not perfection)

- Verified eligible sources and meaningful fields for a scoped test.
- Named usable sample with actual counts, exclusions, source hashes and coverage indicators.
- Coordinate/time/identity mapping sufficient for *that* test, with transparent abstention for unreliable states.

- At least one defensible simple baseline can run; unresolved optional issues explicitly deferred.
- Human-readable feasibility matrix with owner approval of the first bounded experiment.

Stop: do not audit the pose dataset exhaustively or invent complex value/training choices before the passing sample supports them. If necessary pivot to geometry-only or narrow to pass feasibility; don't conceal failures.

06 — FEASIBILITY MATRIX

Original: FEASIBILITY_MATRIX.md

Feasibility decision matrix — fill from observed eligible files

Status: unassessed. Do not convert documentation claims into observations.

Needed capability	Evidence to inspect	Verified observation / denominators	Decision
All eligible resources located	Match/file inventory and official revision	NOT CHECKED	Unknown
Ball carrier / receivers identified	Possession IDs, lineups and ball observations	NOT CHECKED	Unknown
Passing route geometry	Coordinates, pitch size, frames	NOT CHECKED	Unknown
Time-based interception	Continuous frames, speeds, extrapolation	NOT CHECKED	Unknown
Pressing episode selection	Event and phase fields, timestamps	NOT CHECKED	Unknown
Action danger / attacking value	Provider metric definitions, eligible observations	NOT CHECKED	Unknown
Independent evaluation evidence	Observed outcomes and independent baselines	NOT CHECKED	Unknown
Rights to publish match-derived displays	Competition terms and provider permissions	NOT CHECKED	Unknown

Record **PASS / LIMITED / BLOCK / DEFER** with reasons only after the audit. Different model choices require different levels of evidence.

07 — RESEARCH INTEGRITY PROTOCOL

Original: RESEARCH_INTEGRITY_PROTOCOL.md

PressLens — Research Integrity Protocol (proposed v0.1)

Created 2026-11-02. **Status: proposed operating standard for the new researcher-led PressLens project.** This is not evidence that any local repository has been changed. It is intended to be copied into the ChatGPT Project sources or the new research repository. It does not supersede competition rules.

Non-negotiable principle

No paper, author, publication, DOI, dataset statistic, formula attribution, numerical result, or claim of novelty is accepted from LLM recollection alone. A *real citation with a false characterization* is also an integrity failure.

Evidence taxonomy

- **P — Primary-verified:** Original journal publisher, conference proceedings, university repository, original paper, or the official data/competition source confirms the identity and the precise claim being cited. This does not mean the authors' empirical findings were independently reproduced.
- **A — Author-reported:** A genuine primary publication makes an empirical claim, but we have not independently replicated it. Attribute explicitly to the authors.
- **S — Secondary-only / pending:** A credible secondary index reports the work, but original record/full text could not yet be inspected. May inform reading priorities; do not rely on it for decisive claims.
- **H — Hypothesis / proposal:** Our proposed model, expected outcome, or mathematical interpretation. Never present as established evidence.
- **U — Unverified:** Cannot find a reliable traceable source or factual support. Exclude from the paper or label explicitly as unresolved.
- **R — Reproduced here:** A result independently produced with our approved SkillCorner sample, code version, parameters, split, and checks. Always retain provenance and denominators.

Mandatory citation validation

Before citing a work, record: (1) exact title and authors, (2) year and publication type, (3) DOI or stable primary URL, (4) original source actually opened/inspected, (5) precise supported claim and section, (6) whether the result is peer-reviewed, conference/competition, preprint, or thesis, (7) dataset and limitations, (8) whether its code/data/weights can legally be reused, and (9) our evidence status.

For a headline number: verify population/sample, metric, denominator, comparator and evaluation split where accessible; write "authors report" until replicated. If evidence is ambiguous, remove or qualify the number rather than speculate. Never cite a source that merely *cites* the target source as proof the target's result is correct.

Reproducibility and data integrity

- Only SkillCorner competition-eligible match data may be used for the competition analysis/training, unless official rules explicitly allow an additional source. **Published literature is allowed to inform methodology; its empirical training data are not automatically eligible.**
- Inspect third-party packages, pretrained parameters, model weights, bundled training CSVs, licenses and provider restrictions before reuse. Permission to reuse code is distinct from permission to use extra training data.
- Separate software correctness, predictive/calibration performance, construct validity, tactical usefulness, and causal attribution. Never infer one from another.
- Never invent experiment runs, observations, labels, results, sample counts or validation outcomes. Record code version, data provenance, eligibility filters, observation counts and uncertainty.
- Distinguish observed state change from defender-attributable counterfactual estimates; defender counterfactuals are model-based, not proven causal effects.
- Maintain a negative-evidence register: incompatible assumptions, failed tests, bad examples, uncertainty, and methodological limitations must remain visible.

Work process and researcher oversight

1. Researcher describes football question in natural language and approves its interpretation.
2. Research collaborator reviews directly relevant primary literature and provides 2–3 methodological alternatives with assumptions and failure modes.
3. Researcher selects the **bounded** next experiment and success/failure conditions before implementation.
4. Code/analysis is run against verified eligible data; actual outputs and failed cases are shown on the pitch where meaningful.
5. Researcher reviews evidence; accept, revise, defer, or abandon. Update a single concise research checkpoint. No automatic gate advancement.

Anti-loop rule

Further citation searches or audits must identify an unresolved decision they could change, the minimum new evidence required, and a stopping point. Do not keep expanding a literature review because further references might exist. Reopen only for material conflicts, substantive novelty challenges, or newly discovered evidence.

Current state

Source records are in `VERIFIED_SOURCE_REGISTER.csv`. The companion `FOCUSED_LITERATURE_REVIEW.md` contains literature-based assessment **only**. No PressLens model has been trained or evaluated in the new clean-slate project, and no previous Codex result is inherited as validated evidence.

08 — FOCUSED LITERATURE REVIEW

Original: FOCUSED_LITERATURE_REVIEW.md

PressLens — Focused Literature and Method Feasibility Review (v0.1)

Prepared: 2026-11-02. **This is a source-grounded design recommendation, not an empirical experiment.** Nothing has been run on the new PressLens SkillCorner sample. Source status distinguishes publication existence from independent replication.

The question being evaluated

Can a limited-data, reproducible, press-specific model quantify how the *valuable feasible actions* available to the attacking team change through a pressing episode, with a justified extension to attributing off-ball defender contributions?

The owner's football principles: prioritize feasible **danger** of remaining options over option counts or turnovers; track time, space, and opportunity jointly; account for on-ball and off-ball defenders; acknowledge that containment can buy time for support. Scope excludes general transition defence initially.

Verified relevant prior work

Source	Publication status and contribution	Key limit for our application
Fernández, Bornn & Cervone (2021), DOI 10.1007/s10994-021-05989-6	Original peer-reviewed *Machine Learning* article; decomposed EPV into pass, drive, shot, outcome and selection components; trained on proprietary tracking; authors report calibrated components.	Reproduction and training data requirements are high. Using published model ideas is different from legally using pretrained weights/data.
Bischofberger & Baca (2026), DOI 10.1186/s40537-026-01387-8	Original peer-reviewed *Journal of Big Data* article; physically simulated low passes, estimated completion maps, valued accessible space, frame-level off-ball changes and defender-with/without counterfactuals. Authors validated on 3 open matches and published `accessible-space`.	This already covers the *concept* of off-ball defender option suppression! Low/high-pass limitations, tracking error, fitted-model transfer, action selection, and realistic counterfactuals matter.
Everett et al. (2025), DOI 10.1109/DSAA65442.2025.11248003	Listed in original IEEE DSAA technical program and author publication record; GAPP predicts receivers, proposes individual defender influence, authors report tests on 306 EPL matches.	Requires considerable training data; attention-weight-based interpretation is not automatically causal defender credit.
Kim et al. (2026), Sloan Sports Analytics Conference; arXiv:2512.10355	Verified official Sloan paper page + public DEFCON code; uses multiple graph neural components to estimate EPV and defender responsibility.	Ajax training data proprietary; code also references externally sourced Wyscout shot-training features. Not plug-and-play for competition-eligible data.

Source	Publication status and contribution	Key limit for our application
Arbués-Sangüesa et al. (2020), DOI 10.1109/CVPRW50498.2020.00451	Original IEEE/CVF workshop proceedings; body orientation and geometry for passing feasibility; authors report >0.7 top-3 receiver accuracy on >6,000 events.	Top-3 receiver prediction is not calibrated pass completion probability; pose limited to two competition matches.
Everett et al. (2025), DOI 10.1609/aaai.v39i22.34482	Original AAAI proceedings; defensive spatial teamwork optimization under uncertainty; author-reported threat reduction in their data/model.	Multi-agent MDP + Monte Carlo search is too broad for first competition prototype; headline results do not transfer to SkillCorner.
Bojinov & Bornn (2016), MIT Sloan paper	Official Sloan research record; disruptive passing/defensive spatial surfaces.	Prior art for defensive disruption; not equivalent to demonstrated instantaneous off-ball pressure attribution.

Secondary-only lead: Teglund (2026), *Modeling Defensive Influence on Passing Decisions in Football Using Graph Attention Networks*, described as an Uppsala University thesis proposing Pass Suppression Value. A secondary thesis index identifies author/year and abstract, but the original university record/full text was not opened successfully here. Treat as S until primary record is inspected; do not rely on its numerical results as verified.

Methodological options compared

Option A — Static geometry/proximity baseline

- **Inputs:** Eligible XY frames, player/ball positions, team/possession, simple distance and lane geometry.
- **Strength:** Easy to understand; robust comparison baseline; inexpensive.
- **Weakness:** Doesn't properly model time to reach ball, receiver motion, or combined risk-reward. Low ceiling as final contribution.
- **Recommendation:** Mandatory simple baseline, not core contribution.

Option B — Physics-based feasible pass landscape + explicit danger weighting

- **Inputs:** XY, estimated velocities from checked frames, ball carrier and possible receivers, plausible pass trajectories, pitch-position/value function.
- **Strength:** Can evaluate hypothetical/unselected options without fitting a large action-selection neural network; mature published precedent in DAS; interpretable visualization.
- **Weakness:** Motion parameters, high passes, carry opportunities, baseline value scale, motion estimation errors, and selection are nontrivial; published fitted parameters may be based on disallowed external matches.
- **Recommendation: Preferred first candidate** for bounded SkillCorner prototype, subject to permission and input checks. Develop a transparent simple version before adopting any third-party fitted parameters.

Option C — Decision-weighted EPV / graph neural defender attribution

- **Inputs:** Rich event labels, substantial training examples, models for action selection, success and post-action values, credible defender counterfactuals.
- **Strength:** Can capture realistic action choice and interactions if well trained and validated.

- **Weakness:** High sample/computational and verification demands; training data/weights may be ineligible; credit may depend on brittle counterfactuals.
- **Recommendation:** Literature benchmark and optional future extension; **not** first implementation.

Potential distinct contribution (hypothesis, not a novelty claim)

A **press-specific, temporally evaluated** opportunity landscape, with explicit tracking-quality abstentions, comparisons against simple baselines, coach-readable displays of surviving options, and a **conditional** defender-attribution experiment. The individual 'defender suppresses hypothetical pass' concept is already in DAS, GAPP, DEFCON and PSV (secondary-confirmed thesis). Do **not** claim it is new.

Proposed minimum viable measurement (not yet approved)

1. Select only episodes with verified press/possession identity and sufficient tracking coverage.
2. Compute a small set of eligible direct ground-pass opportunities and a modest carry proxy. State clearly what remains outside the action set.
3. For each frame, estimate *success feasibility* and *attacking value on success/failure*, flagging uncalibrated quantities as **scores**, not probabilities.
4. Compare at least two aggregations: maximum surviving action value vs an alternative sensitive to several good choices. Avoid summing mutually exclusive options naively. Selection-weighted expectation requires defensible choice probabilities.
5. Track the score over episode time and compare against a simple geometry/proximity model. **Do not label all drops as caused by the press.**
6. Test sanity/invariance cases, observed pass outcomes where labels support them, errors and abstentions; show a selection of real football pitch episodes to the human researcher.
7. Only then test a modest with/without defender counterfactual, clearly labelled model-dependent and potentially unrealistic. Do not automatically sum credit across interacting defenders.

Specific feasibility and compliance questions

- Official PySport Cup 2.0 rules permit only SkillCorner event/prior-Cup datasets and properly licensed pre-existing code. They prohibit additional match/video/training data. Before adopting `accessible-space`, assess whether default fitted parameters/weights implicitly use outside-match fitting; seek clarification if rules ambiguous.
- SkillCorner lists 20 XY tracking matches, 10 with dynamic events, two with body pose at 25fps vs XY 10fps. **Do not assume** all are clean or usable. Audit only the bounded subset needed for the prototype.
- Stage/press definitions and engagement-event timing need verification against current provider specification and actual data; dataset labels are not independent tactical annotations.
- Coordinate missingness, extrapolation, velocities and possession disagreements can undermine pass feasibility. Decide abstention conditions before evaluating models.
- Expected threat value weights need an eligible, defensible foundation. If merely manually defined/heuristic, label as a heuristic and perform sensitivity tests; do not claim calibrated xT/EPV.
- No independent human tactical labels currently exist in the clean-slate project; pitch review is qualitative, not a standalone statistical accuracy test.

Recommendation and stop condition

Advance to a bounded feasibility specification, not coding an entire EPV/attribution stack: choose one or two press sequences from a verified eligible game, a minimal pass/carry action set, a physics-grounded feasibility calculation and a transparently labeled danger heuristic. Agree on what counts as model failure and abstention before implementation. Preserve the basketball deadline by deferring pose and complex GNN models.

This literature review is sufficient for *selecting a prototype direction*, **not** for asserting a unique contribution or claiming published findings have been replicated. Revisit the search only if a new primary study materially challenges our design/novelty or if a required modeling parameter lacks evidence.

Primary reference URLs

- https://link.springer.com/article/10.1007/s10994-021-05989-6
- https://link.springer.com/article/10.1186/s40537-026-01387-8
- https://doi.org/10.1109/DSAA65442.2025.11248003
- https://dsaa.ieee.org/2025/technical-program/
- https://www.sloansportsconference.com/research-papers/better-prevent-than-tackle-valuing-defense-in-soccer-based-on-graph-neural-networks
- https://arxiv.org/abs/2512.10355
- https://github.com/hyunsungkim-ds/defcon
- https://openaccess.thecvf.com/content_CVPRW_2020/html/w53/Arbues-Sanguesa_Using_Players_Body-Orientation_to_Model_Pass_Feasibility_in_Soccer_CVPRW_2020_paper.html
- https://ojs.aaai.org/index.php/AAAI/article/view/34482
- https://www.sloansportsconference.com/research-papers/the-pressing-game-optimal-defensive-disruption-in-soccer
- https://github.com/jonas-bischofberger/accessible-space
- https://pysport.org/analytics-cup/editions/analytics-cup2
- https://skillcorner.github.io/opendata/
- https://www.uppsatser.se/uppsats/8fa42d2163/ (secondary-only thesis lead)

09 — SOURCE REGISTER

Original: VERIFIED_SOURCE_REGISTER.csv

Source 1

authors_or_org: Fernandez Bornn Cervone
year: 2021
type: Journal article, Machine Learning
doi_or_id: 10.1007/s10994-021-05989-6
primary_or_record_url: https://link.springer.com/article/10.1007/s10994-021-05989-6
evidence_status: P+A
supported_use: Decomposed EPV pass/drive/shot; authors claim learned/calibrated components
limitations_or_actions: Proprietary tracking; model not reproduced here

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Source 2

authors_or_org: Bischofberger Baca
year: 2026
type: Journal article, Journal of Big Data
doi_or_id: 10.1186/s40537-026-01387-8
primary_or_record_url: https://link.springer.com/article/10.1186/s40537-026-01387-8
evidence_status: P+A
supported_use: Physics-based pass completion and DAS; hypothetical defender counterfactual discussed
limitations_or_actions: Limited to simulated low passes; only 3 open matches in authors evaluation

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Source 3

authors_or_org: Everett Beal Matthews Norman Ramchurn
year: 2025
type: Conference, IEEE DSAA
doi_or_id: 10.1109/DSAA65442.2025.11248003
primary_or_record_url: https://dsaa.ieee.org/2025/technical-program/
evidence_status: P+A
supported_use: GAPP off-ball defensive influence and receiver prediction
limitations_or_actions: Original IEEE paper page JS-blocked; checked official program and author record; findings author-reported

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Source 4

authors_or_org: Kim Seo Choi Boomstra Yoon Park
year: 2026
type: Conference, MIT Sloan
doi_or_id: arXiv 2512.10355
primary_or_record_url: https://www.sloansportsconference.com/research-papers/better-prevent-than-tackle-valuing-defense-in-soccer-based-on-graph-neural-networks
evidence_status: P+A
supported_use: DEFCON estimates defending value via expected attacking options
limitations_or_actions: AFC Ajax private match data; published code includes Wyscout features

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Source 5

authors_or_org: Arbues-Sanguesa Martin Fernandez Ballester Haro
year: 2020
type: Conference workshop, IEEE CVPRW
doi_or_id: 10.1109/CVPRW50498.2020.00451
primary_or_record_url: https://openaccess.thecvf.com/content_CVPRW_2020/html/w53/Arbues-Sanguesa_Using_Players_Body-Orientation_to_Model_Pass_Feasibility_in_Soccer_CVPRW_2020_paper.html
evidence_status: P+A
supported_use: Orientation in receiver feasibility
limitations_or_actions: Top3 accuracy is not success probability; does not transfer directly

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Source 6

authors_or_org: Everett Beal Matthews Norman Ramchurn
year: 2025
type: Conference, AAAI
doi_or_id: 10.1609/aaai.v39i22.34482
primary_or_record_url: https://ojs.aaai.org/index.php/AAAI/article/view/34482
evidence_status: P+A
supported_use: Multi-agent defensive teamwork optimization
limitations_or_actions: Author-reported metrics only; different dataset and optimization target

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Source 7

authors_or_org: Bojinov Bornn
year: 2016
type: Conference paper, MIT Sloan
primary_or_record_url: https://www.sloansportsconference.com/research-papers/the-pressing-game-optimal-defensive-disruption-in-soccer
evidence_status: P
supported_use: Defensive disruption spatial surface
limitations_or_actions: Old/event-based baseline; no independent replication

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Source 8

authors_or_org: Teglund Gustav
year: 2026
type: University thesis (secondary index only)
primary_or_record_url: https://www.uppsatser.se/uppsats/8fa42d2163/
evidence_status: S
supported_use: Secondary summary describes Pass Suppression Value using SkillCorner
limitations_or_actions: Primary Uppsala record/full text not accessed; numerical findings pending confirmation

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Source 9

authors_or_org: PySport
year: 2026
type: Official competition rules
primary_or_record_url: https://pysport.org/analytics-cup/editions/analytics-cup2
evidence_status: P
supported_use: Eligible dataset restriction, deadline, licensing
limitations_or_actions: Live rules can change; check before final submission

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Source 10

authors_or_org: SkillCorner
year: 2026
type: Official dataset documentation
primary_or_record_url: https://skillcorner.github.io/opendata/
evidence_status: P
supported_use: Tracking schema, pose caveats, match data overview
limitations_or_actions: Not independent accuracy validation
